Practice Problem Set : p–n Junction Diodes, MOSCAPs, and MOSFETs

EE2104 — Semiconductor Device Fundamentals
Maximum Marks: 20 (Will be scaled for final Grading)

Section 1: p–n Junction Diodes

Question 1

An abrupt silicon p–n junction at $300K$ has $N_A = 2 \times 10^{17} cm^{-3}$, $N_D = 5 \times 10^{16} cm^{-3}$, and $n_i = 1 \times 10^{10} cm^{-3}$.

(a) Calculate the built-in potential V_{bi}, equilibrium depletion width W_0, depletion widths x_p and x_n, and peak electric-field magnitude E_{max}.

(b) Calculate the total depletion width and peak electric-field magnitude at $V = +0.50V$ and $V = -5.0V$. Use the depletion approximation.

(c) For each of the three bias conditions (equilibrium, $+0.50V$, and $-5.0V$), calculate the depletion charge magnitude per unit area on each side and verify charge neutrality.

Question 2

A silicon diode at $300K$ has $p_{n0} = 2 \times 10^{4} cm^{-3}$, $n_{p0} = 5 \times 10^{3} cm^{-3}$, $D_p = 10 cm^2/s$, $L_p = 10\ \mu m$, $D_n = 25 cm^2/s$, $L_n = 50\ \mu m$ and junction area $A = 1 \times 10^{-4} cm^2$. Assume long quasi-neutral regions and ideal low-level injection.

(a) Calculate the minority-carrier lifetimes τ_p and τ_n using $L_p^2 = D_p \tau_p$ and $L_n^2 = D_n \tau_n$. Plot/sketch the minority carrier profiles.

(b) Calculate the reverse saturation current I_0 using the ideal diffusion model.

(c) Calculate the diode current at $V = +0.30V$, $+0.60V$, and $-2.0V$.

Question 3

An abrupt silicon p–n junction at $300K$ has $N_A = 1 \times 10^{16} cm^{-3}$, $N_D = 4 \times 10^{16} cm^{-3}$, $n_i = 1 \times 10^{10} cm^{-3}$, junction area $A = 1 \times 10^{-4} cm^2$, and SRH generation lifetime is $10\ \mu s$

(a) Calculate V_{bi}, the depletion width W, and junction capacitance $C_j = \frac{\epsilon_s A}{W}$ at zero bias and at reverse biases of $-2V$ and $-8V$.

(b) At reverse bias $-8V$, calculate the depletion-region generation-current density using $J_{gen} = \frac{q n_i W}{2\tau_0}$, and calculate the corresponding generation current $I_{gen} = A J_{gen}$.

(c) A measured abrupt-junction C–V plot gives a slope magnitude $S = 2.0 \times 10^{23} F^{-2} \cdot V^{-1}$ For junction area $A = 1 \times 10^{-4} cm^2$, calculate the net doping concentration using $S = \frac{2}{q\epsilon_s A^2 N_{eff}}$, where $N_{eff} = \frac{N_A N_D}{N_A + N_D}$.

Section 2 MOSCAPs

Question 4

An ideal n-channel MOS capacitor has a p-type silicon substrate with $N_A = 1 \times 10^{16} cm^{-3}$, oxide thickness $t_{ox} = 10nm$, and flat-band voltage $V_{fb} = 0V$. The temperature is $300K$.

(a) Calculate the bulk potential $\phi_{Bp} = V_T \ln(N_A / n_i)$ and the strong-inversion surface potential $\psi_{s,th} = 2\phi_{Bp}$.

(b) Calculate the oxide capacitance per unit area $C_{ox} = \frac{\epsilon_{ox}}{t_{ox}}$.

(c) Calculate the maximum depletion width and the corresponding maximum depletion charge magnitude per unit area.

(d) Calculate the threshold voltage. Then calculate the inversion charge magnitude per unit area at $V_g = V_{th} + 0.40\,V$

Question 5

A MOS capacitor has gate area $A = 1 \times 10^{-4} cm^2$. At accumulation, its measured capacitance is $C_{acc} = 1\,nF$. The oxide is SiO_2. At a depletion bias, the measured semiconductor depletion capacitance is $C_{dep} = 0.040nF$, and the corresponding surface potential is $\psi_s = 0.30V$. The measured flat-band voltage is $V_{fb} = -0.25V$.

- **(a)** Calculate the oxide capacitance per unit area $C_{ox} = \frac{C_{acc}}{A}$ and extract the oxide thickness $t_{ox} = \frac{\epsilon_{ox}}{C_{ox}}$.
- **(b)** Using $C_{dep} = \frac{\epsilon_{si} A}{W_d}$, calculate the depletion width W_d at the stated depletion bias.
- **(c)** Using W_d and ψ_s, calculate the substrate doping concentration from $\psi_s = \frac{qN_A W_d^2}{2\epsilon_{si}}$.
- **(d)** Assuming the flat-band shift is entirely due to uniform effective oxide charge, calculate the effective oxide charge density Q_{ox}/A from $V_{fb} = -\frac{Q_{ox}}{C_{ox}}$, and express the result in C/cm^2 and elementary charges/cm^2.

Section 3: (MOSFETs)

Question 6

An n-channel silicon MOSFET operating at $300K$ has a p-type substrate with doping concentration $N_A = 1 \times 10^{16} cm^{-3}$, oxide thickness $t_{ox} = 10nm$, flat-band voltage $V_{fb} = 0V$, channel length , channel width , and effective electron mobility $\mu_{n,eff} = 500 cm^2/(Vs)$, W/L ~ 10

- **(a)** Calculate the oxide capacitance per unit area C_{ox}, the zero-bias bulk potential ϕ_{Bp}, the zero-bias depletion layer capacitance per unit area the zero-bias threshold voltage V_{th0}, and the body factor $m = 1 + \frac{C_{dep0}}{C_{ox}}$.
- **(b)** For an applied gate voltage $V_{gs} = 2.50V$, calculate the drain saturation voltage $V_{ds,sat} = \frac{V_{gs} - V_{th0}}{m}$ and the saturation drain current $I_{ds,sat} = \frac{W}{2mL}\mu_{n,eff} C_{ox}(V_{gs} - V_{th0})^2$.
- **(c)** Calculate the linear-region drain current I_{ds} at $V_{gs} = 2.50V$ and small drain bias $V_{ds} = 0.10V$ using $I_{ds} = \frac{W}{L}\mu_{n,eff} C_{ox}\left[(V_{gs} - V_{th0})V_{ds} - \frac{m}{2}V_{ds}^2\right]$, and calculate the transconductance $g_m = \frac{\partial I_{ds}}{\partial V_{gs}}$ at $V_{ds} = 0.10V$.

Question 7

Subthreshold Swing & OFF-State Leakage Current:

For the n-channel MOSFET in Question 6 with body factor $m = 1.1$ at $T = 300K$:

1. Calculate the ideal subthreshold swing
2. If interface states introduce an interface trap capacitance $C_{it} = 8 \times 10^{-8} F/cm^2$, calculate the degraded body factor and the corresponding degraded subthreshold swing $SS_{degraded} = \ln(10) V_T m_{deg}$.
3. Assuming a subthreshold drain current at $V_{gs} = V_{th0} = 0.83\ V$ of $10^{-6}\ A$, calculate the OFF-state leakage current I_{off} at $V_{gs} = 0\ V$ for both the ideal and degraded cases